

QuickSort Algorithm

3. arr = [-5, 4, 2, 619, 11, 5, 620, -3]

0 1 2 3 4 5 6 7

$$\text{midpoint} = \text{lowIdx} + \left(\frac{\text{highIdx} - \text{lowIdx}}{2} \right) = 0 + \left(\frac{7-0}{2} \right) = 3.5 = 3$$

pivot = arr[3] = 619

S₁ = [-5, 4, 2, 11, 5, -3]

p = 619

S₂ = [620]

S₁ = [-5, -3] p = 2

S₂ = [4, 11, 5]

S₂ = [4, 5, 11]

[-5, -3, 2, 4, 5, 11]

concat

[-5, -3, 2, 4, 5, 11, 619, 620]